Status	Finished
Started	Thursday, 29 October 2026, 9:14 PM
Completed	Thursday, 29 October 2026, 9:18 PM
Duration	4 mins 1 sec
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Binary search ✓
c. Linear search
d. Sequential search

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓
b. When the search algorithm finishes
c. When the list contains duplicate elements
d. When the list is sorted

Question **3**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓
b. At the beginning of the list
c. In the left sublist
d. In the middle of the list

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O👎 ✓
c. O(log n)
d. O(1)

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Quick search
b. Bubble search
c. Merge search
d. Linear search ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised
b. The last element is returned
c. The first element is returned
d. The search is considered unsuccessful ✓

Question **7**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Selection

c. Complexity

d. Bubble ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(1)
- b. O👎
- c. O(log n) ✓
- d. O(n log n)

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the left sublist ✓
- b. In the entire list
- c. At the beginning of the list
- d. In the right sublist

Question **10**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 1000
- b. 100
- c. 10 ✓
- d. 500

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By using a hash function
- c. By dividing the list length by two ✓
- d. By comparing each element sequentially

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search is more efficient than binary search on large lists.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search requires the list to be sorted.

Question **13**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 90 and 94
c. 90 and 99 ✓
d. 89 and 94

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 90 and 99 ✓
c. 94 and 99
d. 89 and 94

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search works on unsorted lists
b. Binary search can find multiple instances of the target element
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list